Case Studies

VM Migration Success Stories

Real-world migration scenarios across industries — from VMware and cloud to KVM/KubeVirt using hyper2kvm + hypersdk.

Case Study #1

Financial Services — VMware Exit

Banking & Insurance | 350 VMs | 12 physical hosts | Multi-datacenter

350	$1.2M	6 weeks	99.7%
VMs migrated	Annual savings	Migration time	First-boot success

Before (VMware)

- vSphere 7 Enterprise Plus — $1.4M/yr
- vSAN + NSX licenses — $300K/yr
- 3 VMware-certified admins
- Broadcom threatening 3x increase
- Windows + Linux VMs mixed

After (KVM + KubeVirt)

- RHEL + KVM — $96K/yr (RHEL subscriptions)
- Local storage + OVS — $0 licensing
- Same 3 admins retrained (2 weeks)
- No vendor lock-in risk
- Critical VMs on libvirt, dev on KubeVirt

Migration Approach

Phase 1: 50 dev/test VMs (1 week). Phase 2: 100 staging VMs (2 weeks). Phase 3: 200 production VMs in maintenance windows over 3 weeks. hyper2kvm batch queue processed 30-40 VMs per night. Windows SQL Server VMs required VirtIO injection — 100% automated.

> *"We expected 2-3 months of migration pain. hyper2kvm's batch automation and auto-boot-verification cut it to 6 weeks. Zero customer-facing downtime."*

Case Study #2

Healthcare Network — Cloud Repatriation

Hospital Network | 80 VMs | Azure | HIPAA-regulated

80	$720K	4 weeks	100%
VMs repatriated	Annual savings	Migration time	HIPAA compliant

Before (Azure)

- 80 VMs on Azure — $85K/month
- Data egress for imaging — $8K/month
- HIPAA compliance concerns (data residency)
- Latency issues for radiology workstations
- Unpredictable monthly bills

After (On-Prem KVM)

- 3 Dell R750 servers — $45K one-time
- Zero egress fees — data stays on-site
- Full HIPAA compliance — local processing
- Sub-millisecond access for PACS/radiology
- Fixed, predictable monthly costs

Key Challenge: HIPAA

Patient data (PHI) couldn't be processed by third-party tools. hyper2kvm runs entirely on-premises with zero cloud dependencies. Audit logs provide the compliance trail required for HIPAA auditors. VHD exports from Azure processed locally with no data leaving the hospital network.

> *"Our HIPAA auditor was thrilled. All migration processing stays on our network. The audit trail from hyper2kvm's structured logging was exactly what we needed for documentation."*

Case Study #3

Manufacturing — Edge Deployment

Automotive Parts | 45 factory sites | SCADA/MES VMs | Remote locations

45	$450K	8 weeks	0
Sites migrated	Annual savings	Fleet rollout	On-site visits

Before (VMware per-site)

- ESXi at each plant — $10K/yr/site
- Total licensing: $450K/year
- On-site IT for updates — $200K/year travel
- No central management
- Inconsistent configurations across plants

After (K3s + KubeVirt)

- Intel NUC at each plant — $1,500 one-time
- K3s + KubeVirt — $0/yr software
- GitOps management from HQ — zero travel
- ArgoCD syncs all 45 sites from one repo
- Identical configs, version-controlled

Key Innovation: Zero-Touch Rollout

Converted all SCADA/MES VMs at HQ using hyper2kvm. Shipped pre-configured NUCs to each factory. On-site electricians plugged them in. K3s auto-joined the fleet via WireGuard VPN. ArgoCD deployed the VM workloads automatically. No IT travel required for any of the 45 sites.

> *"We shipped a box, the electrician plugged it in, and 20 minutes later we could see the factory floor VMs running in our central dashboard. From 45 VMware licenses to zero."*

Case Study #4

Government Agency — Air-Gapped Migration

Federal Government | 120 VMs | Classified network | FedRAMP/FISMA

120 VMs migrated	**$800K** Annual savings	**10 weeks** Migration time	**100%** Air-gapped

Before (VMware on classified net)

- vSphere on isolated SCIF network
- VMware licenses $800K/yr
- License renewal = procurement nightmare
- No internet = no cloud migration tools
- Broadcom vendor risk unacceptable

After (KVM on classified net)

- RHEL + KVM on same hardware
- $0 hypervisor licensing
- hyper2kvm installed from RPM via USB
- All processing offline, no external calls
- No vendor dependency for critical infra

Key Challenge: Complete Air Gap

Classified network has zero internet connectivity. hyper2kvm was installed from RPM packages transferred via approved media. VMDKs exported to encrypted external drives. Conversion performed entirely offline. No telemetry, no phone-home, no cloud APIs called. Full SBOM provided for security review.

> *"Most migration tools assume internet access. hyper2kvm is the only tool we found that works completely offline with zero external dependencies. Our security team approved it in 2 days after reviewing the open-source code."*

Results Summary

Across all case studies.

Metric	Case 1: Finance	Case 2: Healthcare	Case 3: Manufacturing	Case 4: Government
VMs Migrated	350	80	225 (45 sites x 5)	120
Annual Savings	**$1.2M**	**$720K**	**$450K**	**$800K**
Migration Time	6 weeks	4 weeks	8 weeks	10 weeks
First-Boot Success	99.7%	100%	99.5%	100%
Downtime	Zero (customer-facing)	4 hours total	Zero (plant floor)	Planned windows
Target Platform	KVM + KubeVirt	KVM/libvirt	K3s + KubeVirt	KVM/libvirt
Key Compliance	SOC 2, PCI	HIPAA	ISO 27001	FedRAMP, FISMA

775
Total VMs migrated

$3.17M
Combined annual savings

99.8%
Average first-boot success rate

Licensed*
hyper2kvm license